

MERGE-Net: DIFFERENTIABLE SPATIAL ROUTING WITH A PHYSICAL TOKEN BOTTLENECK

Anonymous authors

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ABSTRACT

Vision transformers retain fine image detail but pay to process every patch through every layer. Token merging reduces this cost only when a trainable merging policy also produces the shorter tensor used by downstream blocks. We introduce MergeNet, a local-to-latent vision transformer that separates these roles: continuous spatial mass routing learns where information should flow, and an exact-budget gather physically reduces N patches to K carriers before global reasoning. Routing is restricted to the original image grid, but similarities, transported mass, and carrier locations remain learned. A threshold-based soft top- k relaxation trains the routing scores, while a straight-through carrier weight connects them to the hard gather. Selected mass is retained as a logarithmic key bias; we also give an exact rank-one query/key augmentation that implements this bias with an unmodified FlashAttention kernel. On ImageNet-1K with DeiT-S/8, the trained checkpoint reaches 81.368% top-1 at $2\times$ carrier compression, versus 82.246% for the dense model; matched-budget evaluation gives 81.360% for MergeNet and 81.932% for post-training ToMe. Radius-3 routing improves a matched 150-epoch control by 1.670 percentage points over global routing and 0.598 over flattened windows, and three-seed CIFAR-100 controls retain the spatial advantage after matching receiver degree. The remaining [pending: DTEM score], [pending: latency], and [pending: memory] cells are explicitly reserved for one common-protocol checkpoint evaluation.

1 INTRODUCTION

Fine patch grids preserve object boundaries and small visual details, while global self-attention integrates evidence across an image. Their cost grows with the token sequence. Token merging can consolidate redundant patches instead of discarding them (Bolya et al., 2023), but a learned merger must solve two coupled problems: decide where information should flow, and expose later layers to the physically shorter sequence that realizes the savings.

Continuous relaxations make routing trainable. DTEM learns merge embeddings through soft grouping and merging (Lee & Hong, 2024), but its continuous path retains all token slots; its end-to-end protocol alternates relaxed merge updates with hard token-reducing backbone updates. We ask whether one forward path can learn continuous routing while making the downstream model train on the exact token budget used at inference.

MergeNet answers this question with a local-to-latent architecture. A full-grid encoder performs soft mass transport, then an exact-budget gather changes the shape from N patches to K carriers. Residual cross-attention recovers evidence from the full-grid local features, and global reasoning occurs only on the K carriers. Routing is restricted to receiver positions within radius R on the original two-dimensional grid. This is a weak spatial prior: it excludes implausible long jumps but leaves edge weights, mass, and carrier locations learned.

Two details connect the formulation to efficient execution. We use the threshold-adjusted smooth top- k construction of Su (2024) and couple it to hard indices with a straight-through carrier weight. The gather indices are discrete; the claim is gradient flow to their continuous scores, not differentiation of the indices themselves. We also preserve a carrier’s accumulated mass as a $\log m$ key bias. Because this bias is rank one across queries, an augmented query/key dimension realizes it through an unmodified FlashAttention call (Dao et al., 2022). The released operator implements this

path, while the reported ImageNet checkpoints use PyTorch fused scaled-dot-product attention; we evaluate the two claims separately.

Our evidence supports the routing mechanism while keeping the deployment claim bounded. Spatial routing outperforms global and flattened controls on ImageNet and two selectors across three CIFAR-100 seeds; a degree-matched receiver permutation shows that locality contributes beyond candidate cardinality. The trained 300-epoch model, however, trails dense DeiT-S/8 and post-training ToMe in the available matched-budget accuracy records. We therefore do not claim a state-of-the-art frontier before common-protocol checkpoint timing and the matched DTEM row are available.

Our contributions are:

- a continuous spatial-routing path followed by an exact physical bottleneck, so the latent encoder trains and executes on a fixed carrier budget;
- an original-grid routing prior whose gain is tested against global, flattened-window, selector, and degree-matched controls; and
- mass-aware attention with an exact FlashAttention-compatible reparameterization, together with an evaluation that separates accuracy, token count, arithmetic cost, and measured latency.

2 RELATED WORK

Token reduction. DynamicViT learns token retention (Rao et al., 2021), while A-ViT assigns adaptive depth through halting (Yin et al., 2022). ToMe uses bipartite soft matching and can compress pretrained or jointly trained transformers (Bolya et al., 2023); PiToMe (Tran et al., 2024), MCTF (Lee et al., 2024), and SAD-TM (Xie et al., 2026) modify its matching signal or merge schedule. DiffRate treats layerwise budgets as another learned variable (Chen et al., 2023). Equal final counts therefore do not imply equal layerwise computation.

Differentiable merging. DTEM is the closest predecessor: it decouples the merge representation from backbone features and trains it through relaxed grouping (Lee & Hong, 2024). MergeNet builds on its OpenToMe lineage (Westlake-AI, 2025), but places continuous mass transport and a hard fixed-budget gather in one forward path. The soft top- k relaxation follows Su (2024); it is not claimed as a new selection operator. Proportional attention using token size is also established by ToMe. Our systems addition is the rank-one query/key reparameterization for retaining that bias in FlashAttention.

Spatial structure and recovery. DSM and ALGM combine local and global merging (Heo et al., 2024; Norouzi et al., 2024); GTP-ViT propagates tokens over spatial and semantic edges (Xu et al., 2024). Recent reducers constrain matching along rows/columns or after space-filling reordering (Gong & Lis, 2025; Li & Xu, 2025). MergeNet instead constrains eligible routes by Euclidean radius on the original grid. LookupViT uses cross-attention between compressed and high-resolution streams (Koner et al., 2024); our recovery read is an architectural component rather than a claimed general recovery method. Since local merging can reduce arithmetic without improving throughput (Heo et al., 2024), we keep analytic cost and measured runtime distinct.

3 DIFFERENTIABLE SPATIAL ROUTING AND PHYSICAL COMPRESSION

3.1 LOCAL-TO-LATENT ARCHITECTURE

For patch size p , let $N = (H/p)(W/p)$ be the number of patch slots, excluding CLS. The main model uses $p = 8$, width 384, six heads, six local blocks, and six latent blocks. At 224^2 , all local blocks process 784 patches plus CLS. The local encoder first produces $H^{(1)}, \dots, H^{(6)}$; six transport steps then update the final slots using a metric derived from the corresponding local layer. Transport remains soft and retains all slots. Only the subsequent hard gather reduces the sequence. Figure 1 follows the final architecture in the collaborator deck and the executed classifier. The deck also depicts a generic local-decoder branch; in the executed ImageNet classifier, this branch is instantiated by the recovery cross-attention below and terminates in the CLS classification head.

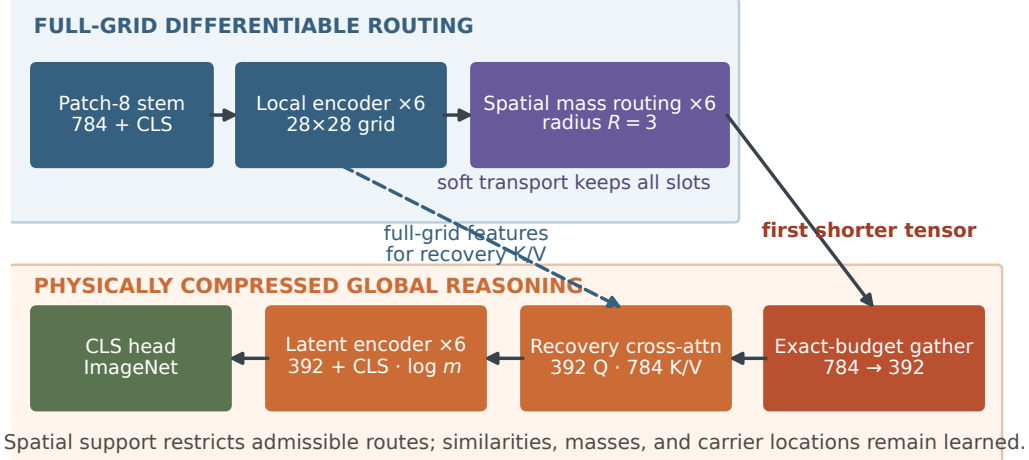


Figure 1: MergeNet at 224 pixels. Soft spatial transport keeps the full grid; the exact-budget gather is the first physically shorter tensor. Recovery reads the final full-grid local features, after which global reasoning uses 392 patch carriers. Counts exclude CLS.

3.2 SPATIAL MASS ROUTING

Index patch i by its original-grid coordinate u_i . At routing step ℓ , donor and receiver sets A_ℓ, B_ℓ are disjoint, and donor i may route only to

$$\mathcal{E}_i^{(\ell)}(R) = \{j \in B_\ell : \|u_i - u_j\|_2 \leq R\}. \quad (1)$$

The main model uses $R = 3$. Coordinates stay tied to original slots throughout routing. The global control removes this constraint; the flattened control uses a width-8 window in row-major sequence order.

A learned $384 \rightarrow 64$ projection produces normalized metrics z_i . On eligible edges,

$$s_{ij} = \langle z_i, z_j \rangle / \tau, \quad q_{ij} = \frac{\exp s_{ij}}{\sum_{r \in \mathcal{E}_i} \exp s_{ir}}, \quad a_{ij} = g_i q_{ij}, \quad (2)$$

where g_i is a soft donor decision. Following Su (2024), a monotone relaxation applies a data-dependent threshold θ after sorting,

$$g_i = F(\alpha \bar{e}_i - \theta), \quad \sum_i g_i \approx k_\ell, \quad (3)$$

with row energy $e_i = \sum_j \exp s_{ij}$, $\alpha = 30$, and the scheduled donor budget k_ℓ . This replaces serial top- k decisions with a parallel soft mask; it does not make the later gather indices differentiable.

Each slot carries mass m_i , initialized to one. A transport step preserves a donor feature and transfers a fraction of its mass into receiver features:

$$\begin{aligned} m'_i &= m_i(1 - \sum_j a_{ij}), & X'_i &= X_i, \\ m'_j &= m_j + \sum_i a_{ij}m_i, & X'_j &= \frac{m_j X_j + \sum_i a_{ij}m_i X_i}{m'_j}. \end{aligned} \quad (4)$$

The implementation clamps residual fractions and denominators. Sparse neighbor lists restrict scored edges, but the released initialization still constructs a dense adjacency buffer; our claim concerns learned routing support, not linear memory for the whole implementation.

3.3 EXACT BUDGET, RECOVERY, AND MASS-AWARE ATTENTION

For compression factor λ , the carrier count is

$$K = N - \lfloor N(\lambda - 1)/\lambda \rfloor. \quad (5)$$

The model hard-gathers the K largest-mass patches. At 224^2 and $\lambda = 2$, $N = 784$ and $K = 392$. A straight-through factor $1 + g - \text{stopgrad}(g)$ on selected features sends gradients to the continuous scores while hard indices execute the budget. Selected carriers Q then recover full-grid evidence through

$$Y = Q + \text{CrossAttn}(Q, H^{(6)}, H^{(6)}), \quad (6)$$

and six latent blocks process K patches plus CLS.

3.4 MEMORY-EFFICIENT RANK-ONE BIASED ATTENTION

Carrier mass enters latent attention as a per-key proportional-attention bias. For N_q queries and N_k keys, let $\beta_j = \log m_j$ and $B = \mathbf{1}_{N_q} \beta^\top$. Then

$$\text{Attn}(Q, K, V; m) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_h}} + B\right)V. \quad (7)$$

The bias is rank one across query positions. Define

$$\hat{Q} = [Q, \sqrt{d_h} \mathbf{1}_{N_q}], \quad \hat{K} = [K, \beta]. \quad (8)$$

Using the *original* softmax scale $1/\sqrt{d_h}$ gives

$$\frac{\hat{Q}\hat{K}^\top}{\sqrt{d_h}} = \frac{QK^\top}{\sqrt{d_h}} + \mathbf{1}_{N_q} \beta^\top, \quad (9)$$

so a standard attention kernel realizes the biased logits without materializing B . The explicit scale is essential: using a kernel default based on the augmented or padded width would not be equivalent.

The released operator pads $d_h + 1$ to $d_{\text{phys}} = 8\lceil (d_h + 1)/8 \rceil$, zero-fills the extra query/key channels, pads V to the same physical width, invokes unmodified FlashAttention, and truncates its output back to d_h . This preserves FlashAttention’s linear activation-memory dependence on sequence length for fixed head width; it does not change the quadratic score computation of global attention. Its generic inference path reuses cached padded workspaces to avoid repeated allocation but still copies Q, K, V into them. Training allocates fresh workspaces for autograd safety, and the caller-padded path clones caller-owned storage before writing the auxiliary channel. We therefore make no zero-copy or peak-memory claim without a separate profile.

The reported ImageNet checkpoints implement Equation (7) through PyTorch fused scaled-dot-product attention with an additive mask. The augmented FlashAttention operator has forward/backward parity tests, but was not their latent-attention path and is not credited with their runtime.

4 EXPERIMENTAL PROTOCOL

Training and evaluation. We train on ImageNet-1K and report EMA top-1 on its 50,000-image validation split. Main 224-pixel runs use AdamW for 150 or 300 epochs with global batch 1,024, cosine decay, five warmup epochs, weight decay 0.05, label smoothing 0.1, Mixup 0.8, CutMix 1.0, RandAugment, random erasing 0.25, and no teacher. The matched geometry controls use learning rate 5×10^{-4} ; the selected MergeNet configuration uses 7.5×10^{-4} . Thus geometry comparisons are recipe matched, whereas the final dense–MergeNet comparison uses each selected configuration. Full arguments and the run inventory are in Appendix A.

The 384-pixel models fine-tune the 300-epoch EMA weights for 30 epochs; the 512-pixel models add 15 epochs. All retain effective global batch 1,024 via gradient accumulation. We report both the best and final EMA values because best-checkpoint selection uses the same validation split. A supplied packet contains 21 complete and two interrupted ImageNet schedules. We recompute endpoints by header name from 3,365 contiguous summary rows and cross-check 3,264 rows against supplied logs; missing-log rows remain summary-only. The company checkpoints were not transferred, so this is a record audit rather than an independent checkpoint replay.

Table 1: ImageNet-1K at 224 pixels and a 392-patch target. Accuracy is percent. Blank cells are requested measurements, not zero values or estimates.

Method	Compression path	Input→final	×	Top-1 (%)	Median ms	Peak GiB
DeiT-S/8	dense, trained	784 → 784	1.00	82.246	—	—
DTEM	matched run	784 → 392	2.00	—	—	—
DeiT-S/8 + ToMe	post-training	784 → 392	2.00	81.932	—	—
MergeNet ($R = 3$)	end-to-end	784 → 392	2.00	81.360	—	—

Table 2: ImageNet routing controls. All rows use the same 150-epoch recipe; accuracies are single-seed EMA top-1.

Routing / probe	LR (10^{-4})	Best (%)	Final (%)
Global	5.0	77.218	77.216
Flat window 8	5.0	78.290	78.278
Spatial $R = 2$	5.0	78.942	78.942
Spatial $R = 3$	5.0	78.888	78.888
Spatial $R = 3$	7.5	79.316	79.316

Compression comparisons. The main table fixes 392 final patches at 224 pixels. ToMe and PiT-oMe are applied after training to the dense checkpoint, while MergeNet is trained with its native bottleneck. The budget sweep evaluates the complete validation set at five budgets. These rows compare deployable checkpoints; they do not isolate the routing rule under identical training. A separate CIFAR-100 study trains two selectors $\times$ four routing geometries $\times$ three seeds for 200 epochs. Its preregistered endpoint is epoch-199 EMA top-1.

Timing. Historical 224- and 384-pixel latency records use synthetic inputs and random weights; one later ToMe sweep also disables proportional attention. We report these only as architecture diagnostics. The primary table leaves latency and memory blank until dense, matched DTEM, ToMe, and MergeNet are measured from trained checkpoints with identical preprocessing, precision, batch, attention policy, token hooks, and CUDA-event timing. The same collaborator packet will supply two inference-only component scores and passive routing traces; no checkpoint transfer is requested.

5 RESULTS

5.1 PRIMARY MATCHED-BUDGET COMPARISON

Table 1 fixes the final budget at 392 patches, excluding CLS. The MergeNet row is the 50,000-image re-evaluation (81.360%); its training summary reports a best EMA value of 81.368%. Dense DeiT-S/8 reaches 82.246%, and post-training ToMe reaches 81.932%. Thus the current evidence establishes a learnable physical bottleneck, but not a stronger accuracy frontier. The matched DTEM score and common-protocol systems cells remain explicit blanks.

5.2 DOES THE SPATIAL PRIOR HELP?

The 150-epoch ImageNet controls isolate routing support at learning rate 5×10^{-4} . Radius 3 reaches 78.888%, improving over global routing by 1.670 points and flattened windows by 0.598 (Table 2). Radius 2 is only 0.054 points higher, so one seed does not identify an optimal radius. The global arm is an architectural control with MergeNet’s local/latent split, not a reproduction of DTEM.

Radius changes both geometry and candidate count. We therefore repeat the comparison on CIFAR-100 with a receiver permutation that preserves each realized receiver degree. Across three seeds, radius 3 ranks first under both the historical batch-normalized threshold selector and a corrected per-sample selector (Table 3). Every one of the 18 paired radius-3 minus control differences is positive. Against the degree-matched control, the mean gains are 1.42 ± 0.30 and 1.19 ± 0.13 points, respectively. These runs support a locality effect beyond degree, but do not transfer ImageNet uncertainty.

Table 3: CIFAR-100 epoch-199 EMA top-1 (%), mean $\pm$ sample standard deviation over seeds 42, 43, and 44.

Selector	Global	Flat 8	Degree-matched	$R = 3$
Historical selector	68.48 $\pm$ 0.10	69.29 $\pm$ 0.54	69.17 $\pm$ 0.62	70.59 $\pm$ 0.88
Rowwise selector	68.77 $\pm$ 0.32	69.64 $\pm$ 0.80	69.21 $\pm$ 0.75	70.41 $\pm$ 0.67

Table 4: Claim-to-ablation map. “Done” rows contain complete experiments.

Claim	Control	Result
2-D routing	global, flat, $R = 2/3$	Done; Table 2
Locality beyond degree	receiver permutation	+1.42 $\pm$ 0.30 points
Selector dependence	historical vs per-sample	$R = 3$ first under both
Budget behavior	20 matched token settings	ToMe higher in 20/20
Mass-aware attention	set log m bias to zero	[pending: score]
Full-grid recovery	bypass cross-attention	[pending: score]

Table 4 ties each mechanism claim to a control. The two remaining component interventions require only one inference pass of the collaborator-held final checkpoint; they are diagnostic ablations rather than separately trained architectures.

5.3 LONG SCHEDULES, RESOLUTION, AND BUDGETS

At 300 epochs, MergeNet trails dense DeiT-S/8 by 0.878 points at 224 pixels (Table 5). Fine-tuning narrows this gap to 0.394 at 384 pixels; at 512 pixels the gap is 0.450, so the three points do not support a monotonic convergence claim. Different MergeNet radii are used across this ladder and the 512-pixel arm descends from the radius-scaled 384 checkpoint.

Changing only the evaluation-time carrier budget gives MergeNet top-1 values of 81.380, 81.360, 81.074, 80.850, and 80.228% at 523, 392, 314, 262, and 196 patches. ToMe applied to the dense checkpoint is higher at every one of these 224-pixel budgets, with gaps from 0.572 to 1.524 points; the same ordering holds at all five 384-pixel budgets. This is a clear negative result for the current end-to-end model. It is not a controlled comparison of training objectives, because ToMe starts from the stronger dense checkpoint. Detailed curves, optimizer probes, and complete schedules are in the appendix.

6 EXECUTION EVIDENCE

A code-derived accounting estimates 14.13 GMAC per image for MergeNet and 22.41 GMAC for dense DeiT-S/8 at 224 pixels, a 36.9% reduction. This includes projections, MLPs, attention, routing transport, and recovery, but omits sorting, indexing, and data movement. Since all six local blocks remain dense and recovery costs $O(NK)$, a $2\times$ final carrier reduction does not imply a $2\times$ model reduction.

Existing runtime records confirm this distinction. On one H20-3e with random weights, fp16, synthetic batch 64, and a guarded CUDA-event harness, MergeNet takes 45.71 ms versus 42.68 ms for dense DeiT-S/8 at 224 pixels and uses more peak allocated memory. At 384 pixels it takes 136.54 versus 170.61 ms, a 20.0% reduction. A local A100 reproduction does not recover this crossover. These architecture benchmarks are useful boundary checks, but they cannot be paired with trained-checkpoint accuracy. The primary table therefore reserves common-protocol timing rather than reporting an unsupported Pareto frontier.

The inherited soft threshold routine also normalizes its score range across the batch. Holding one sample fixed while changing another alters its soft weights and can produce cross-sample gradients; a constant batch can be non-finite without a denominator clamp. The three-seed study repeats the routing comparison with a per-sample selector and preserves the ordering, but the main ImageNet checkpoints were trained with the historical operator. We report this implementation boundary in-

Table 5: Best EMA top-1 across resolution schedules. The gap is MergeNet minus dense in percentage points.

Pixels	Schedule	DeiT-S/8 (%)	MergeNet (%)	Gap (pp)
224	300 scratch	82.246	81.368	-0.878
384	300+30	82.976	82.582	-0.394
512	300+30+15	83.068	82.618	-0.450

stead of silently attributing corrected operator behavior to those checkpoints. Reproduction details, parity tests, and batch probes are in Appendix C.

7 DISCUSSION AND LIMITATIONS

The experiments separate a positive mechanism result from a negative system result. Original-grid support improves over global, flattened, and degree-matched controls, so the spatial prior is useful within this architecture. The complete model nevertheless remains below dense DeiT-S/8 and ToMe in recorded top-1, and it is slower at 224 pixels in the available random-weight benchmark. The current evidence therefore supports “trainable spatial routing with a physical bottleneck,” not a state-of-the-art accuracy–efficiency frontier.

The design also retains substantial full-grid work. Soft transport does not shorten the local stage; recovery reads all local features; and sparse scoring still incurs indexing and initialization costs. A local edge constraint does not bound the final receptive field after multiple transports, nor does it establish semantic boundary preservation. The 150-epoch ImageNet geometry study has one seed, while the replicated degree-matched evidence is on CIFAR-100. Finally, validation-set checkpoint selection makes reported best scores selected rather than independent estimates.

Two collaborator-side artifacts remain. One numeric packet fills the matched DTEM, latency, memory, mass-bias, and recovery cells under a single protocol. One passive trace from the final checkpoint supplies routing visualizations with checks that tracing leaves logits, top- K indices, and mass conservation unchanged. No checkpoint transfer is required. Until those artifacts arrive, the placeholders stay visibly unfilled and make no contribution to the reported results.

8 CONCLUSION

MergeNet combines continuous spatial mass transport with an exact-budget gather, so a learned routing policy feeds a physically shorter latent sequence in the same forward path. The original-grid prior consistently improves routing controls, including a degree-matched three-seed study, and accumulated carrier mass admits an exact FlashAttention-compatible key-bias construction. Current ImageNet results also expose the remaining gap: the model does not yet exceed dense DeiT-S/8 or post-training ToMe in accuracy, and its deployment frontier awaits common-protocol checkpoint measurements. This evidence supports the compression boundary and spatial-routing mechanism while defining a precise, testable path to a stronger system result.

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A TRAINING RECIPE AND EVIDENCE INVENTORY

Table 6 summarizes the resolved run arguments. The base width is 384 with six heads and MLP expansion four. Stochastic depth is configured at 0.1; MergeNet constructs separate local and latent ramps, unlike a single twelve-block dense ramp. The global and flattened routing controls use their dense/historical scoring paths, while radius-2/3 routing uses the sparse spatial backend; the routing support and its numerical implementation should therefore both be recorded. EMA validation replaces raw-model metrics before the summary row is written, so `eval_top1` is an EMA metric.

Setting	224-pixel scratch	384/512-pixel fine-tune
Epochs	150 or 300	30 at 384; 15 at 512
Effective global batch	1024	1024
Optimizer	AdamW	AdamW
Learning rate	5×10^{-4} or 7.5×10^{-4}	10^{-5}
Minimum learning rate	10^{-5}	10^{-6}
Weight decay	0.05	10^{-8}
Warmup epochs	5 (probe: 20)	1
Warmup learning rate	10^{-6}	10^{-6}
EMA decay	0.99996	0.99996
Mixup / CutMix	0.8 / 1.0	0 / 0
Label smoothing	0.1	0.1
Random erasing	0.25	0.25
RandAugment	<code>rand-m9-mstd0.5-inc1</code>	same
Gradient clip norm	1.0	1.0
Validation crop percentage	0.9	1.0
MergeNet auxiliary-logit weight	ramp to 0.05	0.05
Teacher / feature / routing distillation	disabled	disabled

Table 6: Main recipe. Launch records take precedence over configuration-file defaults, especially for microbatch size and gradient accumulation.

The 150-epoch MergeNet runs start the auxiliary-logit contribution at epoch 10 and ramp it over 10 epochs; the 300-epoch run uses epoch 20 and a 20-epoch ramp. Fine-tunes enable weight 0.05 from the start. The recorded environment is Python 3.10.20, PyTorch 2.7.1+cu118, torchvision 0.22.1+cu118, timm 0.9.11, FlashAttention 2.7.3 and Triton 3.3.1 on H20-3e GPUs. Both four and eight ranks divide 50,000, avoiding validation-sampler padding for this split. Other world sizes require a separate check.

Three nominal launcher failures occurred after complete metric histories: the 150-epoch DeiT-S/16, the compound $\lambda = 4$ run, and dense 512-pixel fine-tuning. Complete summaries and supplied checkpoint receipts distinguish them from early failures. Receipts record checks on another machine, not weights independently loaded in this audit. The λ curriculum and 512-pixel MergeNet fine-tune are now complete 150-epoch and 15-epoch records. The weight-decay probe and DeiT-initialized run remain partial.

B DETAILED BUDGET AND TIMING COMPARISONS

Table 8 gives the values behind the main accuracy panels. The MergeNet checkpoint uses radius 3 at 224 pixels and radius 5.143 after 384-pixel fine-tuning. Final patch budgets match within a row, but layerwise attention and MLP inputs do not. ToMe and PiToMe shorten sequences within successive blocks, while MergeNet gathers once between stages. The companion CSV also contains the two resolution-transfer panels using 224-pixel checkpoints without fine-tuning.

At the approximately half-size operating point, ToMe processes 1,151 patches and MergeNet 1,152, taking 129.810 and 155.255 ms respectively. These times do not prove a joint ordering for the accuracy sweep’s ToMe setting. A common harness should use exactly 1,152 patches with proportional attention enabled, verify the checkpoint and preprocessing, and remove shape hooks before timing. An earlier latency benchmark silently clamped several requested ToMe budgets to the same achieved count; those obsolete cells are excluded.

Run ID	Epochs	Status	Best top-1
s2_deit_p16_150e	150/150	Complete	77.196
s2_deit_p8_150e	150/150	Complete	79.992
s2_mn_flat_w8_150e	150/150	Complete	78.290
s2_mn_global_150e	150/150	Complete	77.218
s2_mn_l4_150e	150/150	Complete	78.366
s2_mn_r2_150e	150/150	Complete	78.942
s2_mn_r3_150e	150/150	Complete	78.888
s2_mn_r3_lr75_150e	150/150	Complete	79.316
s3_deit_p16_300e	300/300	Complete	79.692
s3_deit_p8_300e	300/300	Complete	82.246
s3_mn_r3_lr75_300e	300/300	Complete	81.368
s3_deit_p8_ft384_30e	30/30	Complete	82.976
s3_mn_r3_ft384_30e	30/30	Complete	82.582
s3_mn_r5_ft384_30e	30/30	Complete	82.574
s4_mn_lamcurr_150e	150/150	Complete	79.310
s4_mn_warm20_150e	150/150	Complete	79.238
s4_deit_ft512_15e	15/15	Complete	83.068
s4_mn_ft512_15e	15/15	Complete	82.618
s5_mn_dp07_150e	150/150	Complete	79.534
s5_mn_lr1e3_150e	150/150	Complete	79.350
s5_mn_wd03_150e	140/150	Partial	79.054
s6_deitB_p16_300e	300/300	Complete	80.442
s7_mn_deitinit_150e	105/150	Partial	79.206

Table 7: Status in the September 19 packet. Partial metrics are included for accounting only and are not endpoints. The DeiT-B best score is a pre-overfitting peak (Appendix D). The companion data retain full precision and source summary lines.

Checkpoint / pixels	Patches	MergeNet	ToMe	PiToMe	Δ ToMe
224 trained / 224	523	81.380	82.038	81.876	+0.658
224 trained / 224	392	81.360	81.932	81.742	+0.572
224 trained / 224	314	81.074	81.814	81.560	+0.740
224 trained / 224	262	80.850	81.830	81.500	+0.980
224 trained / 224	196	80.228	81.752	81.206	+1.524
384 tuned / 384	1537	82.500	82.868	82.658	+0.368
384 tuned / 384	1152	82.562	82.866	82.534	+0.304
384 tuned / 384	923	82.230	82.800	82.452	+0.570
384 tuned / 384	770	82.002	82.792	82.334	+0.790
384 tuned / 384	576	81.300	82.742	82.296	+1.442

Table 8: Top-1 percentages at equal final patch counts; Δ is ToMe minus MergeNet in percentage points. These values cannot be paired with Table 9 to construct an accuracy–latency frontier.

Separate 384-pixel radius-3 benchmarks remain distinct from the radius-5.143 sweep. Speed percentages at different batch sizes, radii or harness settings should not be averaged into an unexplained range. Peak reserved memory and peak allocated memory also measure different quantities.

C OPERATOR-LEVEL DIAGNOSTIC AND REPRODUCTION BOUNDARIES

The batch-dependence probe imports only the released threshold-selection function. It does not instantiate an ImageNet model or load a checkpoint. A fixed first row is evaluated alone and with controlled second rows; a weighted loss on the first row tests cross-sample gradients. Constant-input cases test finite forward and backward behavior. These probes establish operator properties, not an accuracy gain from replacing the operator.

Per-sample normalization removes the demonstrated coupling, but does not by itself validate the closed-form threshold solver over every budget and tie pattern. A patched closed-form candidate passes 116 of 120 tensor coverage cases and is not recommended for integration. A separate slow

ToMe patches	MN patches	ToMe ms	MN ms	ToMe GiB	MN GiB
1536	1538	147.30	178.17	2.793	3.619
1151	1152	129.81	155.25	2.764	3.564
922	923	122.44	143.02	2.746	3.721
769	770	118.17	136.59	2.738	3.719
575	576	106.45	129.38	2.723	3.557

Table 9: Separate 384-pixel synthetic latency experiment: batch 64, H20-3e, fp16 autocast, random initialization, MergeNet radius 5.143, and ToMe proportional attention disabled. Patch counts exclude the class token here. Memory is peak *reserved* GiB, unlike the historical 224-pixel record, which reports peak allocated memory. Token-counting hooks remain active during timing.

reference finds a per-row threshold t for normalized scores z satisfying

$$\sum_i h(z_i - t) = k, \quad h(\delta) = \begin{cases} 1 - \frac{1}{2} \exp(-\delta), & \delta > 0, \\ \frac{1}{2} \exp(\delta), & \delta \leq 0. \end{cases} \quad (10)$$

For $0 < k < N$, the sum is continuous and strictly decreasing in t . With $v_i = \partial \mathcal{L} / \partial h_i$, implicit differentiation gives

$$\nabla_{z_i} \mathcal{L} = h'_i \left(v_i - \frac{\sum_j v_j h'_j}{\sum_j h'_j} \right). \quad (11)$$

PyTorch then propagates gradients through the per-row normalization to the raw scores. The reference normalizes per row, solves in float64 with 60 bisection steps, and casts the result back to the input dtype. It passes all 120 coverage cases for finite forward/backward values and selection mass, plus three nondegenerate gradient checks. Maximum mass error is 9.16×10^{-5} in float32 and 1.14×10^{-13} in float64; the tested cross-sample output and gradient dependencies are zero. These tests cover random, constant, nearly constant and outlier inputs across $N \in \{4, 16, 64, 392\}$ and several budgets. This is a correctness reference, not a fused production kernel or a trained replacement for the reported model. Finite gradients can still be large near a zero score range. Appendix H subsequently tests GPU/autocast inference with local CIFAR weights. The later campaign in Section 5.2 trains a float32/Triton rowwise implementation, which is distinct from this slow float64 reference. Its completed CIFAR runs do not establish ImageNet training behavior. Mass tolerances in these checks are 5×10^{-4} for float32 and 10^{-6} for float64.

The artifact subset includes unmodified per-run summaries, resolved arguments with host paths removed, launch-protocol extracts, sanitized successful sweep records, audited benchmark tables, and source hashes. The CPU rebuild script recomputes best/final metrics, resolves sweep records by timestamp, checks numeric joins, and regenerates tables and figures. It never launches training. Source and claim ledgers retain their literature retrieval dates; the collaborator-packet audit was refreshed on September 21. Company policy prevents ImageNet weight transfer; independent ImageNet checkpoint re-evaluation is unavailable. The September 19 packet adds completed 512-pixel, curriculum, and hyperparameter runs. The refreshed audit matches 3,264 of 3,365 summary rows to delivered EMA logs, leaving 101 rows without corresponding log records (Section 4). Those rows are not discarded, but retain their summary-only evidence status.

D DEiT-B/16 IS NOT A USABLE SCALE ANCHOR

A 300-epoch DeiT-B/16 run at 224 pixels completes under the same global-batch and seed protocol as the small-scale campaign, with drop-path 0.1 inherited from the DeiT-S configurations. Table 10 reports the peak and final EMA scores. Train loss continues to fall while evaluation loss rises after the peak at epoch 179. The final top-1, 78.566%, is below our DeiT-S/16 result of 79.692% at the same 196-token budget. Published DeiT-B 300-epoch training uses drop-path 0.2. We therefore do not append this run to the DeiT-S/8 table, and we do not launch a paired MergeNet-B comparison from these weights.

Epoch	Train loss	Eval loss	EMA top-1 (%)
150	2.968	0.889	80.060
179 (peak)	2.808	0.895	80.442
299 (final)	2.348	1.095	78.566

Table 10: DeiT-B/16, 300 epochs, drop-path 0.1. Peak and final scores must be reported together; the peak is not a converged result.

E A BOUNDED LAYOUT OPTIMIZATION EXPERIMENT

We additionally test a single LocalAttention module on an A100-SXM4-80GB. The original module constructs query/key/value tensors in batch-head-token-channel order, then transposes and copies them for the FlashAttention wrapper. An isolated variant constructs batch-token-head-channel tensors directly and adjusts the global class-query attention views and final reshape accordingly. It does not alter the released model, load a checkpoint, or apply an optimizer update.

The test uses PyTorch 2.6.0+cu124 and FlashAttention 2.7.4.post1, width 384, six heads, window radius 16, zero attention dropout, float32 parameters and inputs, and fp16 autocast. Four batch-2 parity cases cover sequence lengths 785 and 2,305, global class attention on/off, and query/key normalization. Outputs and input gradients are bitwise equal in these cases. Parameter gradients are also equal except for the normalization case, whose maximum relative L_2 error is 5.77×10^{-7} . This coverage does not establish all-shape or dropout/RNG-preserving equivalence for a complete model.

Batch	Tokens	Mode	Original ms	Layout ms	Reduction range
64	785	Forward	1.5600	1.0448	13.50–34.03%
64	785	Forward + backward	5.1590	4.3999	14.68–25.64%
16	2305	Forward	1.0318	0.9369	6.88–36.68%
16	2305	Forward + backward	5.3740	4.9176	5.45–11.83%

Table 11: Single-module A100 measurements. Times are medians of three round medians, each with five warmups and twenty CUDA-event samples. The final column is the range of paired-round time reductions, not a confidence interval. Tokens include the class token. Backward includes a squared-output loss and gradient clearing, with no optimizer.

Measurement order alternates by round, and the harness requires an initially empty device, establishes its CUDA process identity, then rejects a changed process set at subsequent checks. All iteration samples and snapshots are retained. The observed reductions support this local engineering direction, but the round variation is appreciable. Both module instances are resident, so their recorded memory peaks are not independent per-model memory measurements. These results do not predict the full model’s speedup, validate an HBM-traffic estimate, transfer H20 timings to A100, or resolve the separate ToMe attention-policy mismatch.

F RESERVED TRAINED-CHECKPOINT ROUTING VISUALIZATION

The final ImageNet checkpoint is unavailable locally. The collaborator will export four to six validation examples with the actual donor/receiver partition, nonzero transport edges, token-mass maps, final carriers, and at least one failure case. The trace is passive: it records maximum logit difference with tracing enabled, equality of top- K indices, and mass conservation error after every step. Without a source-membership trace, the figure will show per-step transport rather than an invented final segmentation.

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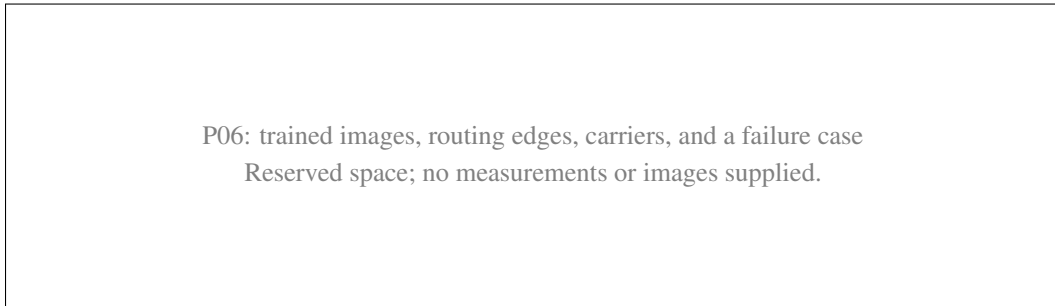


Figure 2: Reserved trained-checkpoint routing visualization. The blank is not a model observation and will be replaced only by a verified trace.

G HISTORICAL CIFAR-100 SPATIAL-ROUTING ABLATION

We retain a historical CIFAR-100 experiment as a limited diagnostic of the routing geometry. The August campaign contains 60 new runs: two MergeNet configurations, five input sizes, and six routing settings (Global and Euclidean radii $R \in \{1, 2, 3, 4, 8\}$), all using seed 42. The configurations have local/latent depths of 6/6 with final compression ratio $\lambda = 2$, or 4/8 with $\lambda = 4$; both use width 384, six heads, and 8×8 patches. Each model was trained from scratch for 200 epochs without distillation. Training used AdamW, learning rate 10^{-3} , weight decay 0.05, a cosine schedule with 20 warmup epochs, effective batch size 200, and EMA decay 0.9998. The primary endpoint is the final epoch-199 EMA top-1 accuracy on the 10,000-image CIFAR-100 test set. The frozen local campaign protocol designated $R = 3$ as the primary setting before the recorded training launches; this is an internal protocol record, not an external preregistration. We keep that setting even though $R = 2$ has a slightly higher mean.

The original images are 32×32 pixels and are resized to the listed input sizes. Thus, this sweep changes the patch grid over upsampled images and does not test native high-resolution image content. Evaluation used FP16 autocast and deterministic fast grouping, with batches of 200, 200, 200, 100, and 50 at sizes 160, 192, 224, 256, and 320, respectively. Global and the finite-radius settings were trained in the same frozen implementation and share the non-routing configuration. The additional Flat reference uses the historical flattened-window implementation with width parameter 8 and comes from ten earlier August runs. Its recorded non-routing training arguments and dependency versions match, but its runtime code snapshot differs; three Flat runs resumed within their original jobs. We therefore use Flat as a historical descriptive reference.

Configuration	Input	Global	Flat [†]	$R = 3$	ΔG	ΔF
6/6, $\lambda = 2$	160	67.80	68.79	69.42	+1.62	+0.63
	192	69.42	69.93	70.85	+1.43	+0.92
	224	69.50	70.53	70.53	+1.03	+0.00
	256	69.50	70.12	71.03	+1.53	+0.91
	320	69.85	70.81	71.42	+1.57	+0.61
4/8, $\lambda = 4$	160	65.29	66.69	67.92	+2.63	+1.23
	192	69.87	69.75	71.27	+1.40	+1.52
	224	68.21	69.87	70.52	+2.31	+0.65
	256	67.35	68.30	69.40	+2.05	+1.10
	320	66.05	67.08	68.83	+2.78	+1.75
Mean		68.284	69.187	70.119	+1.835	+0.932

Table 12: Historical CIFAR-100 final-epoch EMA accuracy (%) and $R = 3$ deltas (percentage points) against Global (ΔG) and Flat (ΔF). Each row uses seed 42; the mean gives equal weight to the ten configuration–input pairs. [†]Flat is a historical reference with a different code snapshot. Input denotes the resized side length of native 32×32 images. These are observations under the stated evaluation batches and historical dense-score implementation.

At the fixed $R = 3$ setting, accuracy exceeds Global in all ten pairs, with a mean improvement of 1.835 percentage points. Relative to historical Flat, nine pairs improve and one ties, with a mean difference of 0.932 points. Across the complete sweep, the mean accuracies for Global and radii 1, 2, 3, 4, and 8 are 68.284, 68.774, 70.225, 70.119, 70.047, and 68.414%, respectively. Radii 2, 3, and 4 each exceed their matched Global control in all ten pairs. These correlated, single-seed observations provide evidence that restricting routing neighborhoods can help in this particular setting; they do not establish statistical significance, native-resolution transfer, or superiority over external token-reduction methods.

The audit reconciled all 60 new endpoints and the ten Flat endpoints with raw 200-epoch summaries, completion receipts, and frozen manifests. Archived full-test evaluations of the 60 new epoch-199 EMA checkpoints show identical generic- and fast-grouping logits; their accuracies differ from the training summaries in 31 cases by at most 0.02 points. The table preserves the registered training-summary endpoint. The initial historical audit checked receipt hashes and checkpoint presence, without freshly evaluating or rehashing all checkpoints. The historical spatial implementation constructs a dense donor–receiver score matrix before masking, so these results do not validate the later sparse backend or its efficiency. Furthermore, both historical implementations use a threshold operator whose normalization range is shared across a batch, allowing a sample’s routing to depend

on its batch companions. Appendix H subsequently rehashes and reevaluates two of these checkpoints and quantifies their batch sensitivity; the full historical sweep remains tied to its recorded evaluation batches. Section 5.2 reports a later 24-run campaign with three seeds, two selectors, and a degree-matched control; those runs are independent trainings and are not substitutes for the table above.

H INDEPENDENT FOLLOW-UP WITH AVAILABLE LOCAL ARTIFACTS

Company policy prevents transfer of the ImageNet checkpoints. We therefore separate two feasible follow-ups: architecture timing using random weights, and prediction tests using existing, locally available CIFAR-100 weights. Neither experiment independently reproduces the company ImageNet scores.

H.1 A COMMON A100 INFERENCE HARNESS

We compare dense DeiT-S/8, ToMe, and the canonical sparse-backend MergeNet on an A100-SXM4-80GB using PyTorch 2.6.0+cu124, timm 0.9.11 and FlashAttention 2.7.4.post1. All cells use batch 64, float32 parameters and inputs, and fp16 autocast. The compressed models retain exactly 392 patches at 224 pixels and 1152 at 384 pixels, excluding CLS. A diagnostic forward records block outputs; all hooks are removed before timing. ToMe source tracing is disabled. Proportional attention is explicitly controlled, with its enabled setting matching the recorded accuracy protocol.

Each cell has ten warm-up iterations and thirty CUDA-event samples in each of three rounds, reversing model order in the middle round. A selected-GPU process guard requires an empty device before setup and only our process during the experiment. Models are allocated and released sequentially. Table 13 reports the median of three round medians. It measures model execution, excluding data loading, backward and updates.

Model	224 px (ms)	384 px (ms)
Dense DeiT-S/8	30.09	109.43
ToMe, proportional attention on	34.42	133.36
ToMe, proportional attention off	29.71	96.05
MergeNet $R = 3$	49.16	144.97
MergeNet $R = 3$, BNHD	46.56	138.11
MergeNet $R = 5.143$	–	218.25
MergeNet $R = 5.143$, BNHD	–	211.40

Table 13: Independent synthetic A100 inference, batch 64. All weights are randomly initialized; these are not trained-checkpoint frontier points. BNHD changes the local-attention tensor layout. The 384-pixel radius-5.143 model has the same final patch budget as radius 3.

On this local hardware/software stack, the original MergeNet is slower than dense DeiT at both resolutions. The BNHD change reduces complete-model radius-3 time by about 5%, considerably less than the isolated-module improvement. The proportional-attention setting materially affects ToMe latency. Thus neither the H20 crossover nor a ToMe timing measured with proportional attention disabled should be treated as accelerator-independent evidence. The A100 comparison changes the hardware and software stack relative to the H20 experiment; it does not isolate a hardware-only cause. No historical accuracy point is attached to these random-weight timings.

H.2 LAYOUT VALIDATION UNDER A COMPLETE TRAINING GRAPH

The layout comparison uses identical weights and images, restored CPU and CUDA random states, fresh per-sample routing, and zero dropout at both 224 and 384 pixels. With the dense routing reference backend, outputs, input gradients and all parameter gradients are bitwise equal. With the sparse backend and an unscaled squared-logit loss, the strict gradient comparison fails; an unchanged-model repeat also fails, so these runs do not isolate an error introduced by BNHD. Their input-gradient relative errors are approximately 4.8% and 16.3%, respectively.

We retain these failed tests and add a fixed loss scale of 1024, unscaling all gradients before comparison. The sparse-backend tests then pass the predefined relative- L_2 limits of 0.002 for outputs and 0.003 for gradients. Maximum parameter-gradient relative errors are below 0.0016; unchanged-model repeat controls are below 0.0018. This diagnoses sensitivity of the unscaled fp16 test and supports bounded layout parity. It does not establish long-run training equivalence, eliminate sparse-kernel nondeterminism, or validate arbitrary dropout and precision settings.

H.3 BATCH SENSITIVITY IN LOCALLY TRAINED CIFAR MODELS

We strictly load the epoch-199 EMA states for the $\lambda = 2$, 224-pixel global and radius-3 models from the August spatial campaign. Fresh file hashes match the completion receipts, and saved model arguments are checked against the frozen protocol. The original dense routing runtime and CIFAR normalization are retained. Every evaluation covers the same 10,000 test images. The batch-200 baseline reproduces radius-3 accuracy exactly and is 0.01 pp below the recorded global endpoint; it is not claimed to reproduce both endpoints bitwise.

Evaluation	Global (%)	$R = 3$ (%)	Gap (pp)	Changed G/R3
Original order, $B = 200$	69.49	70.53	+1.04	0/0
Original order, $B = 64$	69.41	70.55	+1.14	168/35
Permuted order, $B = 200$	69.44	70.52	+1.08	163/28
Rowwise reference, $B = 200$	69.17	70.55	+1.38	548/114

Table 14: Independent local CIFAR EMA evaluation. “Changed” counts the images whose predicted class differs from that model’s original-order batch-200 result. The last row replaces the historical selector by the rowwise bisection reference without updating weights. It is a distinct inference intervention, not the historical model’s score.

Changing batch size or image order changes 163–168 global-model predictions and 28–35 radius-3 predictions, while top-1 changes remain at most 0.08 pp and 0.02 pp, respectively. The radius-3 advantage remains 1.04–1.14 pp in these three protocols. Thus the implementation issue is observable in a trained model, but it does not erase the geometry benefit in this bounded pair of checkpoints.

A separate probe holds the first 16 images fixed in a batch of 64 and replaces only the other 48 images. Maximum logit changes are 0.335938 for global routing and 0.048828 for radius 3; exact same-batch repeats give zero change. Substituting the per-row bisection reference gives zero change in both probes. This intervention supports a selector-level explanation for these particular batch-composition effects. None of the first 16 predicted classes changes in this small probe; full-set class changes are reported separately above.

The full-set selector intervention changes 548 global predictions and 114 radius-3 predictions. Global top-1 falls by 0.32 pp, while radius-3 changes by +0.02 pp. Consequently, restoring per-sample selection is not a general accuracy-preserving substitution for old weights. The reference passes this bounded GPU inference check, but remains an untrained correctness implementation. This checkpoint intervention alone does not assess its learning behavior; the subsequent training experiment is described below.

A subsequent 24-run CIFAR-100 campaign trains both selectors from scratch with three seeds and a degree-matched control (Section 5.2). Those endpoints are new trainings, not re-evaluations of the two August checkpoints in this appendix.

The artifact includes all per-image predicted classes, labels, raw timing samples, observed token shapes and hash receipts. These local tests address specific implementation and comparison questions; two seed-42 CIFAR models cannot establish multi-seed reliability or ImageNet performance after a selector change.